

Analyte Concentration Determination by Using the External Standard and Standard Addition Methods

Prelab reading: Harris, *Quantitative Chemical Analysis: Chapter 4-8: Calibration Curves; Chapter 5*

1. Learning goals

- Understand the impact of sample matrix on quantitative analyses by comparing the external standard method with the standard addition method.
- Measure uncertainty of slope, intercept, and concentration using statistical tools in Excel.
- Determine the limit of detection and limit of quantitation using a calibration curve.

2. Introduction

The external standard method using a calibration curve is one of the most straightforward ways to measure the concentration of an unknown sample. It is called external standard because the standard or known material is separate or external to the unknown material. The calibration curve is so powerful because it demonstrates the sensitivity of the method, which influences limits of detection and quantitation.

The standard addition method is a method for determining the concentration of an analyte in an unknown sample with a complex matrix. The matrix is everything other than the analyte that is present in the sample, and can either increase or decrease observed signal. The standard addition method controls (and minimizes) the effect of the matrix by fixing the amount of sample, while varying the concentration of the analyte by adding a known standard. In short, one fills (usually) five volumetric flasks each with the same volume of sample solution. Then, an increasing volume of standard analyte solution is added to each flask. Finally, each flask is filled to the mark with solvent. By preparing the solutions this way, each flask contains the same concentration/amount of the matrix, and the effect of the matrix on the analyte in the sample is the same in each solution.

We will measure our solutions using spectrophotometry, which is a technique that measures the absorbance of light as a function of the wavelength of light. Absorbance is the log of the ratio of light intensity entering the sample (I_0) to that of the light leaving the sample (I):

$$A = \log \frac{I_0}{I} = \epsilon bc \quad (1)$$

and is equal to the distance that the light travels through the sample (b , unit = cm), the concentration of the species that absorbs light (c , unit = mol L⁻¹), and the molar absorptivity coefficient (ϵ , unit = M⁻¹ cm⁻¹). The molar absorptivity coefficient is an intrinsic property of a species and is a measurement of how strongly that material absorbs light. The molar absorptivity of potassium ferricyanide is 1020 M⁻¹ cm⁻¹ and the path length of the cuvettes is 1 cm. Using this information, the concentration of a species can be directly determined using Eq. 1, which is commonly known as Beer's Law.

In this experiment, you will use the external standard and standard addition methods to determine the concentration of potassium ferricyanide in an unknown solution using spectrophotometry. Ferricyanide, Fe(CN)₆³⁻, is a polyatomic anion that absorbs near-UV and blue light in the range from 350 to 475 nm with a maximum absorbance occurring at 420 nm. This is a two-week experiment. The external standard portion will be run on week one, and the standard addition portion will be run on week two.

3. Reagents

- Reagent grade potassium ferricyanide salt
- Unknown potassium ferricyanide sample

4. Procedures

Instructions for preparation of external standard solutions

- Clean and rinse the volumetric flasks carefully with distilled water. A good method to determine if your glassware is truly clean is to observe if any water droplets are present in the neck of the flask. If there are water drops present, the glassware has not been cleaned properly. Perform one final rinse of each flask with the solvent (water).
- In a 100.00 mL flask, prepare a stock solution of potassium ferricyanide with a concentration of 0.0200 M. Prior to coming to lab, you must calculate the amount of $K_3Fe(CN)_6$ (in grams) that is required to make 100.00 mLs of the 0.02 M solution. Show your calculations in your notebook.
- Weigh the salt $K_3[Fe(CN)_6]$ salt and prepare the solution. Write down the mass of potassium ferricyanide that you used and the concentration of the solution that you prepared. This means that you may not have exactly 0.0200 M but a slightly more or less concentrated solution. However, you should not be off by too much (i.e., 5% of the “true value”). **Calculate the actual concentration of potassium ferricyanide in the stock solution and use this in all of your subsequent calculations.**
- Use a 500.0 mL volumetric flask to prepare a 0.001 M potassium ferricyanide standard (solution #5) by diluting your stock solution. Use the four 50.0 mL volumetric flasks to make standards #1 through #4 by diluting the 0.001 M standard solution (standard #5) with the solvent (0.01 M KOH). Prior to lab you should have the calculations for the preparation of solutions 1-5 calculated.
- Cover the top of each flask with parafilm and carefully invert each flask several times to mix the solutions well.
- **Note:** When preparing the standards, keep in mind that it is better to use accurate and precise volumes rather than trying to get the exact concentrations. You can always calculate the concentrations on the basis of the stock solution and the dilutions that you made. For example, standard #5 may be end up to be 0.00098 M instead of 0.00100 M. That is okay! As long as the 0.00098 M is accurate and precise.

Standard #	Potassium Ferricyanide Concentration (M)
1	0.00020
2	0.00040
3	0.00060
4	0.00080
5	0.00100

Spectrophotometric Measurements for Part 1

Detailed information on performing the spectrophotometric measurements is described in Appendix A

- It is good practice to keep all of your results as you collect them, this includes blanks. Keep track of which spectral traces correspond to which solutions in your lab notebook.

- Determine the wavelength of maximum absorbance for potassium ferricyanide.
 - Place a sample of one of your potassium ferricyanide solutions into your cuvette and measure.
 - There should be a large peak in the region of 420 ± 10 nm. Record the location of this peak in your notebook.
You will make all of your measurements at this wavelength during this experiment.
- Measure the absorbance of the background at least ten times at the maximum absorbance wavelength that you previously measured.
 - After each measurement discard the solution in the cuvette and replace it with fresh solution.
 - In this experiment, the background is 0.01 M NaOH.
 - These measurements should be very similar, within 1-2% of each other. If your variability is larger than this, check the calibration settings on the UV-Vis
 - Record each absorbance measurement in a table in your lab notebook.
- Measure the absorbance of each standard solution of potassium ferricyanide that you prepared three times.
 - After each measurement discard the solution in the cuvette and replace it with fresh solution.
 - Record each absorbance measurement in a table in your lab notebook.
- Measure the absorbance of the unknown solution of potassium ferricyanide that was provided to you at least seven times.
 - After each measurement discard the solution in the cuvette and replace it with fresh solution.
 - Record each absorbance measurement in a table in your lab notebook.
- After you have completed all of your measurements, discard all of your waste in the sink. Clean your glassware with distilled water and return it to the appropriate cupboard.

5. Cleanup

- Clean and rinse all the glassware that you have used and place it on the drying rack. Place volumetric pipettes in the oven (careful: oven is hot!).
- Wipe your bench space clean.
- Dispose of all other wastes in the designated waste container.

6. Data Analysis for Standard Curve

- *Include the calibration curve for the absorbance vs. $K_3Fe(CN)_6$ concentration and the table for the calculation of the unknown concentration in your report.*
- There are two objectives for the data analysis portion of this experiment. The first is to determine the molar absorptivity of potassium ferricyanide from the linear regression of the standard curve and compare with the established “true” value. The second is to use your standard curve to determine the concentration of an unknown solution of potassium ferricyanide.

Analysis of blank measurements

- Measure the average signal and standard deviation of the ten blank measurements.
- *You will need to subtract the signal of the blank from all of your measurements once you’ve determined the average.*

Plot of calibration curve

- Plot the absorbance as a function of concentration of the standards. Make sure that for the standard concentrations, you use the actual values based on the amount of $\text{K}_3[\text{Fe}(\text{CN})_6]$ that you measured out in the beginning of lab. For assistance with plotting data in Excel, see Section 2-11 in Harris.
- Perform a linear regression of the x,y scatter data and show the trend line on the graph.
- Using the LINEST function in Excel, determine the slope, intercept, and uncertainty in the slope and intercept. Report these values clearly in your lab write-up.
- Label this as Figure 1 in your lab report, and provide a description of the figure. The figure caption should provide enough detail to understand the result of the experiment.

Calculations

- Absorbance and concentration are related by Beer's Law, $A = \epsilon bc$, where ϵ is the molar absorptivity ($\text{M}^{-1} \text{cm}^{-1}$), b is the path length that light travels through the solution (cm), and c is concentration ($\text{M} = \text{mol L}^{-1}$). The slope of linear regression of A versus c (in mol L^{-1}) should equal the product of ϵb .
- Calculate the value of ϵ for $\text{K}_3[\text{Fe}(\text{CN})_6]$ using your experimental data. Be sure to provide a measure of uncertainty in the molar absorptivity. Calculate the absolute and relative errors. Report both of these values in your lab report.
- Present the results in a table similar to that displayed below. Pay attention to units and significant figures.

Measured molar absorptivity ($\text{M}^{-1} \text{cm}^{-1}$)	Uncertainty (SD) of molar absorptivity ($\text{M}^{-1} \text{cm}^{-1}$)	RSD of molar absorptivity (%)	Absolute error of molar absorptivity ($\text{M}^{-1} \text{cm}^{-1}$)	Relative error of molar absorptivity ($\text{M}^{-1} \text{cm}^{-1}$)
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- Calculate the concentration of the potassium ferricyanide unknown solution using the average value of the absorbance from the unknown sample and the calibration curve. Also calculate the uncertainty of the concentration and calculate the relative standard deviation (RSD) of concentration.
- Provide all the information in a final table similar to that displayed below. Pay attention to units and significant figures.

Concentration of potassium ferricyanide in unknown sample (M)	Uncertainty (SD) of concentration (M)	RSD of concentration (%)
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- Calculate the limit of detection (LOD) and the limit of quantitation of the method based on your results.

Appendix A: Detailed information on performing spectrophotometric measurements

Safe Handling and proper use of the Vernier LabQuest 2

The Vernier LabQuest module and the SpectroVis Plus spectrophotometer are expensive pieces of equipment. Please take the following precautions while you use them. Treat the device as if it was your cellphone and you had to pay the bill if you damaged it.

A. Treat the devices carefully. Place them on the bench top well away from the edge and do not walk around with them (except for getting a unit and storing them).

B. Keep liquids well away from the module. Organize your bench space such that your solutions are separated from your Vernier module. Do not use the module with non-aqueous solutions. Be extra careful when using acids and bases.

C. Use the stylus to touch the screen, rather than your fingers. You may transfer chemicals when using your fingers.

D. When done, switch off the device. Wipe off any liquids that were accidentally spilled on the module. Put the stylus back in the device. Put everything away in its proper place: LabQuest modules with LabQuest modules in one box, spectrometers with spectrometers in another box, etc.

E. Report any irregularities to your lab instructor.

- For this experiment, we will use a Vernier Spectrometer, an instrument that can quickly measure the entire absorption spectrum of a sample. It is necessary to set the instrument's scale to where no light is absorbed and to where all the light is absorbed. This process is called "calibration".
- Obtain a LabQuest module, SpectroVis Plus spectrophotometer, a power supply, and a USB cable from your lab instructor. The power supply plugs into the LabQuest and the cable connects the spectrophotometer with the LabQuest unit. The LabQuest should also have a small stylus in a slot on the back of the instrument.
- Obtain two plastic cuvettes – only touch the cuvettes on the very top $\frac{1}{4}$ or on the frosted sides. You do not want to get finger prints on the clear windows where the light must travel.
- Plug the USB cable from the spectrophotometer. Turn on the LabQuest unit (red on button on top left). The instrument will automatically take you to the Absorption App (red box with "USB: Abs") if you have your spectrometer plug in before you turn on the console. If you don't see the app started, you will need to start LabQuest App to get to the correct screen.
- When you are in the app -- touch "Sensors" then "Data Collection" on the top of the screen. Enter the following settings:

Sample time: 200 ms

Wavelength smoothing: 3

Samples to average: 6

Wavelength range 380 to 700 nm

Then touch "ok"

- Then go to Sensors > Calibrate > USB: Spectrometer.
- Touch ok or just wait. You will get a new screen that tells you the spectrometer is warming up. Wait for the spectrometer to complete the warm up of the lamp.
- After the instrument warms up, fill one cuvette with 0.01 M KOH, wipe it clean with a Kimwipe, and place it in the sample chamber. Make sure the cuvette is positioned such that the clear windows are aligned with the white arrow and white lamp on the instrument (that is the direction the light will travel). Touch finish calibration on the screen.
- The instrument will take ~ 5 seconds to calibrate itself (it is figuring how the signal from 100% of the light traveling through). When it says “calibration completed”. Touch ok.
- The screen will go back to the original screen in the app (with a multi-colored background). It may also default immediately into a measurement (will see red box at bottom left rather than green arrow). It may also go back to the screen with the red box (“USB: Abs”). If the instrument didn’t start, start a measurement with only the water in the cuvette by touching the green “play” button. Stop the measurement by touching the red “stop” symbol in the lower left. A very noisy flat line centered around 0 should appear on the graph. The magnitude of the noise should not be larger than 0.0070. If it is, you need to recalibrate (see step 10).
- (OPTIONAL) If your line is not centered around 0, you can recalibrate your instrument by touching the Meter icon (looks like an old-fashioned gas gauge) at the upper left on the screen. Then select “Sensors” and “calibrate”. Touch “ok” and you will get back to your original calibration screen shown in step 6.
- If your line in step 9 was flat and centered around 0, you are ready to measure.
- Fill the second cuvette 2/3 full with your standard 5. Place the cuvette into the spectrophotometer such that the clear sides are aligned with the white arrow. Touch the green arrow. You may discard the previous run by touching discard. Within a second, the absorption spectrum of the solution should appear on the screen. Touch the red box to stop the measurement. Note: if you discard the previous data, your new spectrum will be labeled run 1. If you choose to store the previous data, your new spectrum will be labeled run 2. A subsequent run will be named run 3, etc. If you store, you can at any time revisit your runs by touching the run # box.
- Move the line cursor on the graph to the peak of the absorption curve. Record the wavelength (the x value, shown on the right side of the screen, in nm). The x value will be in nm (don’t mistakenly write down the y-value). You will now do all your measurements at that absorbance wavelength.
- Touch the Table icon at the upper right on the screen. You should now get the spectrum values as x,y data. Scroll down the table to your wavelength. Write down the absorbance (Abs) value for your wavelength, this is A_B .
- Remove your cuvette with the standard 5. Discard the solution in the cuvette into a “waste beaker”. Rinse the cuvette three times with the standard 4 and discard into your “waste”. Fill the cuvette 2/3’s full with the standard 4. Place in the correct orientation in the absorption spectrometer.
- Touch the green arrow on the bottom left to take the new spectrum. Tell the console to discard the old data. Wait a few seconds for the new data to show up in the table. Record the absorbance value for your second standard solution (at the same wavelength).

- Repeat the last two steps (step 15 and 16) with each of your standard solutions #3 to # 1 and record an absorbance value for each solution. Be careful to rinse the cuvette each time with the new solution.
- Rinse the cuvette with solvent and three times with the unknown sample solution. Fill it 2/3 with the unknown sample solution. Measure the absorbance of the unknown sample solution at the same wavelength and write down the absorbance in your notebook. Rinse the cuvette and fill 2/3 with new solution of the unknown sample and repeat until you have 5 independent measurements of the absorbance of the unknown sample solution at the same wavelength.